

MIKKYLA DANIELLE DEL ROSARIO URIENZA

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EDUCATION

University of California, Davis

B.S. Electrical Engineering | GPA: 3.8

Honors & Activities: Tau Beta Pi • Dean's Honor List (4x) • E-SEARCH Undergraduate Research

Relevant Coursework: VLSI Design, Integrated Circuits Fabrication, Digital Systems II, Electronic Circuits, Embedded Systems, Computer Architecture, Control Systems, Device Physics, Signals & Systems

Davis, CA

Mar 2026

EXPERIENCE

IT Help Desk - Engineering Department | University of California, Davis

Mar 2023 - Jun 2023

Part-time Technical Support

Davis, CA

- Resolved assigned technical-support tickets independently for engineering department users; diagnosed workstation and user issues and escalated cases requiring additional support.
- Maintained an organized help-desk workspace and communicated task status and unresolved issues to supervising staff.

Undergraduate Researcher | E-SEARCH, UC Davis College of Engineering

Fall 2023

Advisor: Kevin Immanuel Gubbi

Davis, CA

- Modified Verilog modules in an ODIN-inspired neuromorphic architecture to model disrupted calcium homeostasis and Alzheimer's-like neuronal behavior using finite-state-machine and digital-logic changes.
- Developed and synthesized a prototype hardware-Trojan model for spiking-neural-network behavior; explored FPGA implementation on a Basys 3 Artix-7 platform and hardware-security implications for on-chip learning.

SELECTED ENGINEERING PROJECTS

8-bit Barrel Shifter - CMOS/VLSI Design | Cadence Virtuoso

Winter 2026

- Designed transistor-level CMOS building blocks from inverter and logic gates through 2:1 multiplexers to an 8-bit barrel shifter; created schematics, symbols, testbenches, and physical layouts.
- Verified functionality with DC/transient simulation, parametric transistor sizing, DRC, and LVS; analyzed switching thresholds, propagation behavior, and rise/fall-time tradeoffs.

Autonomous Camera-Based Vehicle - Primary Technical Contributor

Winter-Spring 2025

- Designed the vehicle PCB in Altium, integrating H-bridge motor control, voltage regulators, MOSFETs, filtering, OpenMV controller, servo, and high-current routing; implemented camera-guided control in Python.
- Built frame-by-frame lane detection and error/deflection-angle logic with PID-based PWM speed/steering correction; project placed Top 3 and was demonstrated at the UC Davis engineering open house.

Biometric Attendance System - Primary Implementation

Winter 2025

- Implemented an embedded attendance system using a CC3200 microcontroller, UART fingerprint module, IR remote/receiver, display interface, and AWS-backed roster/attendance storage.
- Ported fingerprint-sensor functionality from an Arduino library to C and implemented user enrollment, ID entry, biometric validation, menu navigation, and real-time attendance logging.

Semiconductor Fabrication & Characterization | UC Davis Cleanroom

Winter 2026

- Performed photolithography and wafer-processing steps including spin coating, soft bake, mask alignment/exposure, development, etching/deposition-related process work, and device characterization; evaluated alignment and feature-quality differences across exposure conditions.

TECHNICAL SKILLS

IC / VLSI & Semiconductor: Cadence Virtuoso, CMOS schematic/layout, DRC/LVS, transistor sizing, parametric analysis, photolithography, cleanroom semiconductor fabrication, wafer/device characterization

Digital / Embedded: Verilog, SystemVerilog, FPGA development, Intel Quartus, RISC-V, embedded C, microcontrollers, GPIO, ADC, PWM, UART, SPI

PCB / Circuit Design: Altium Designer, LTspice/Spice, schematic capture, PCB layout/routing, analog/digital circuit analysis

Programming & Lab: Python, C, C++, MATLAB; oscilloscopes, logic analyzers, function generators, probe stations, semiconductor parameter analyzers